

Prep sheet — Khaleel & Al-Nimri 2026: deep bite with bite turbos alone, plus a reverse-curve wire, or plus posterior elastics

Khaleel R, Al-Nimri K. Assessment of three different techniques in correcting deep overbite: a prospective clinical study. Angle Orthod. 2026;96(5):528–535. DOI 10.2319/100925-850.1.

Design: prospective, non-randomised, single-operator, operator-allocated | n: 81 enrolled, 80 completed, 27/27/27 nominal | Groups: I turbos+0.016×0.022 NiTi RCOS; II turbos+0.014 NiTi+box elastics; III turbos+0.014 NiTi | Cephalograms at bonding and after levelling.

30-second opening

"Everyone had bite turbos; does a reverse-curve wire or box elastics speed bite opening? Yes: elastics 44 days, reverse curve 56, turbos alone 94; an auxiliary halves it. The reverse curve cost 7.5° of incisor flaring and 6–9° of molar tip-back; elastics flared 3.5° but opened the mandibular plane most. The operator chose the arms, partly on cooperation, so I take the halving, not the 11-day ranking."

Bottom line

Adding a reverse-curve wire or box elastics to bite turbos cut time to a 30% overbite from 94.15 to 55.72 and 44.42 d (Table 2); the price is 7.46° of proclination (reverse curve) or the largest mandibular-plane opening, +1.84° (elastics). The claimed 1 mm of true intrusion is 0.99 mm at Point I against 0.71 and 0.86 mm without a reverse curve, never tested (Table 4).

Eight numbers to remember

Number	What it is	Say it like this
55.72 ± 10.52 / 44.42 ± 10.88 / 94.15 ± 7.08 d	Days to overbite <30%, I / II / III (Table 2; text prints 55.7 ± 20.5 and 94.4 ± 7.1 d); I vs II 11.3 d, 95% CI 2.32–20.29, P=.010	"Eight, six and thirteen weeks; an auxiliary halves it, and the 11 days between the auxiliaries is under one review interval"
7.46 ± 2.52° / 3.46 ± 3.89° / 1.55 ± 1.39°	Lower incisor to corpus axis, I / II / III (Table 3), P<.001	"The reverse curve flares twice what elastics do and five times turbos alone"
0.99 / 0.71 / 0.86 mm	Point I vertical change, I / II / III (Table 4); no between-group test printed	"Every arm moved Point I about a millimetre, with or without a reverse curve"
8.94 ± 3.45° and 5.99 ± 4.75°	L7 and L6 distal tipping, Group I (Table 3), P<.001 vs II and III	"A cinched reverse curve tips the molars back 6 to 9°; the L7 distal cusp did not extrude (–0.64 mm)"
+1.35 / +1.84 / +0.69°	Mn-Mx change, I / II / III (Table 5); only II vs III differs, P=.011 (FH-Mn P=.026)	"Elastics open the plane most, turbos alone least; the reverse curve differs from neither"
3.92 / 3.65 / 3.57 mm from 5.51 / 5.33 / 5.13 mm	Overbite reduction, I / II / III (Results text; baseline Table 1); no test printed	"Same amount whichever you choose, because the endpoint is a threshold"
81 → 80; 15.82 ± 2.12 / 16.00 ± 2.17 / 17.35 ± 2.92 y	Enrolled, completed (one lost both turbos); mean ages I / II / III (Table 1)	"Adolescents the Discussion calls nongrowing; turbos alone was the oldest arm"
Overbite <30% + Shimstock contact, every 14 d	Correction endpoint and review interval (Methods)	"Duration means time to a threshold, called fortnightly by the unblinded operator"

What each arm did

Arm	How the bite opened	Main side effect	Days
I: turbos + 0.016×0.022 NiTi RCOS, cinched distal to 7s	Incisor tipping and intrusion (tip –2.46 mm; Point I 0.99 mm) plus premolar extrusion 1.05–1.06 mm (Tables 3–4)	Flaring 7.46°, tip 1.01 mm forward; L6/L7 distal tipping 5.99°/8.94°; L4 mesial 4.44°	55.72 ± 10.52
II: turbos + 0.014 NiTi + 3/16-inch medium box elastics	Lower posterior extrusion 0.90–1.19 mm (text: 0.6–1.5 mm) plus upper posterior 0.28–0.42 mm, the only arm with upper extrusion above 0.15 mm (Table 4)	Largest vertical opening: Mn-Mx +1.84°, FH-Mn +1.31°, LFH +1.56% (Table 5); LI-CA +3.46°	44.42 ± 10.88
III: turbos + 0.014 NiTi only	Passive lower posterior eruption 0.86–1.30 mm; incisor tip –1.25 mm; upper incisor –0.43 mm (Table 4)	Slowest; LI-CA +1.55°; Mn-Mx +0.69°	94.15 ± 7.08

Strengths and weaknesses

Strengths

- Prospective; one operator, bracket system and turbo mould; lower arch bonded 7–7.
- Clear endpoint, reviewed every 14 d (monthly in the 2022 RCT, literature).
- Corpus-axis superimposition, Greig's Point I; intraexaminer ICC 0.92–0.98.

Weaknesses

- Operator allocation on alignment and cooperation: Group III oldest (17.35 y), most crowded (1.81 mm, Table 1); Group II compliance-selected.
- Unblinded operator called the endpoint fortnightly, so the 11.3 d gap is inside one visit; no control, no follow-up.
- Standard errors fit arms of about 29/31/20 rather than 27 each, and 14 of 26 turbos-only values match the 2022 RCT's turbo arm (derived; read conflicts.md before raising this).

What it changes at the chair

- An auxiliary cuts bite opening from three months to 6–8 weeks. Caveat: turbos-alone was oldest and most crowded.
- Choose elastics or reverse curve on incisor position and compliance, not the 11.3 d gap (under one visit). Caveat: compliance-selected arm.
- A rectangular NiTi reverse curve as first wire flares incisors 7.46 ± 2.52° even in aligned arches (crowding 1.25 mm). Caveat: Proffit's textbook torque error.
- Not the reverse curve for intrusion: Point I 0.99 vs 0.71 and 0.86 mm without one, untested. Caveat: the 2022 RCT's turbos arm gave 0.28 mm (literature).

- Elastics open the plane most (+1.84°) and are the only arm with upper posterior extrusion above 0.15 mm (0.28–0.42 mm). Caveat: long faces excluded; extrusion suits short, growing faces (Proffit).

Three questions to ask

- "Point I moved 0.99, 0.71 and 0.86 mm, untested between groups; is the millimetre of true intrusion a reverse-curve effect or just turbos on a 0.014 wire?" *Why it lands: their own marker; Group III had no intrusive mechanics.*
- "Elastics beat the reverse curve by 11 days, CI 2 to 20, with 14-day reviews and a compliance-selected arm; is that gap real, or do you choose on 3.5° versus 7.5° of flaring?" *Why it lands: Tables 2 and 3 together; gap under one visit.*
- "Turbos alone extruded the lower posteriors as much as elastics, 1.30 versus 1.12 mm at the first-molar mesial cusp, yet took twice as long; where does the speed come from?" *Why it lands: Table 4 read across arms; only Group II extruded upper posteriors beyond 0.15 mm (0.28–0.42 mm).*

Five questions you will be asked

- Primary outcome?** Days from bonding to overbite <30% with Shimstock contact: 55.72/44.42/94.15 d; undeclared, unregistered.
- Baseline balance?** No P<.05 (Table 1), but Group III oldest (17.35 y), most crowded (1.81 mm); Group I ANB 5.14° despite a 5° ceiling.
- Reverse-curve effects?** LI-CA +7.46°, tip –2.46 mm, 1.01 mm forward, Point I 0.99 mm; L6/L7 +5.99°/+8.94° distal, L4 +4.44° mesial.
- Who gets elastics?** A compliant, short- or average-faced adolescent with forward incisors: 44.42 d, +3.46°; not long faces, Mn-Mx +1.84°.
- Evidence level; practice change?** CEBM Level 3. Add an auxiliary when speed matters, chosen on inclination and compliance; claim neither intrusion nor stability.

Exam radar

Ten facts most likely to be tested

- Non-randomised; operator allocated on alignment, inclination, cooperation.
- Durations (Table 2): 55.72/44.42/94.15 d, all pairs significant; I–II 11.3 d, P=.010.
- Endpoint: overbite <30% plus Shimstock, fortnightly; reduction 3.92/3.65/3.57 mm from 5.51/5.33/5.13 mm, untested.
- LI-CA (Table 3): +7.46°/+3.46°/+1.55°, P<.001.
- Superimposition: corpus axis at Pm (Ricketts), palatal vault (Broadbent); Point I at 0.66 × incisor length from the tip (Greig): 0.99/0.71/0.86 mm, untested.
- Group I (Table 3): L6 5.99°, L7 8.94° distal; L4 4.44° mesial.
- Group II (Table 4): lower extrusion 0.90–1.19 mm; upper 0.28–0.42 mm (other arms ≤0.15 mm).
- Mn-Mx (Table 5): +1.35°/+1.84°/+0.69°; only II–III differ (P=.011; FH-Mn P=.026).
- Ages 15.82/16.00/17.35 y, guardian consent, called nongrowing; no untreated control.
- Inclusion: overbite >40%, ANB 1–5°, average/reduced LFH, ≤3 mm crowding, average smile; overjet >6 mm excluded.

Three likely MCQ stems

- Fastest arm and margin over the reverse curve? A. Reverse curve, 11 d. B. Elastics, 11 d. C. Turbos alone, 39 d. D. Elastics, 50 d. **B:** Table 2, 44.42 vs 55.72 d.
- Proclined lower incisors, 55% overbite: which protocol adds least proclination? A. Turbos+RCOS. B. Turbos+elastics. C. Turbos+0.014 NiTi. D. All equal. **C:** LI-CA +1.55° (Table 3).
- The Conclusions credit the reverse curve with 1 mm of true intrusion. What does Table 4 actually show? A. Only Group I intruded. B. Similar Point I change in all arms, never tested between groups. C. 0.014 NiTi intrudes as well as a reverse curve. D. A group-size artefact. **B:** 0.99/0.71/0.86 mm, untested.

One likely oral question

"17-year-old woman, Class I, 60% overbite, average face, upright incisors, 2 mm crowding, wants speed: which protocol, what warnings?"

She fits the criteria; turbos with box elastics: 44.42 d against 55.72 and 94.15 d (Table 2). Warnings: elastics work only when worn; posteriors extrude about 1 mm; the mandibular plane opens 1.8° (Tables 4–5). Upright incisors could take the reverse curve's 7.46°, but elastics get there with 3.46° and no molar tip-back (Table 3). No follow-up past a 30% overbite: I promise neither total time nor stability; Proffit's bite-plane retainer for years applies.

Before the session

When	Do	Time
Day received	Read summary.md and key-numbers.md; mark the eight numbers	40 min
Next day	Run the flashcards; rehearse the five answers aloud with the numbers	30 min
Day before	This handout only	20 min
In the room	Open with the 30-second version; ask question 1 first	—

Verdict in one line

A reverse-curve wire or box elastics on bite turbos halves time to a 30% overbite in short- or average-faced, well-aligned adolescents; the ranking between them, the millimetre of intrusion and every skeletal contrast sit inside operator-allocation noise.